

Short Analysis Pipeline

BIOLM0051

Archie Benn

Scenario

A suitcase containing several packages of unidentified frozen meat has been intercepted at Heathrow Airport by UK Border Force officers during routine inspections. The shipment had no import documentation, raising concerns that the meat may originate from protected or endangered species.

Given the serious implications for public health, food safety, and wildlife conservation, the case has been referred to the Animal and Plant Health Agency (APHA). As a bioinformatician assisting with the inquiry, you have been tasked with determining the species of origin for each meat sample. DNA barcoding has already been performed in the laboratory, and you have now received the resulting FASTAQ sequence files for analysis.

Abstract

Introduction

Methods

The FASTQ files in this report are small and the use of HPC resources in this case is not necessary. However, this workflow is designed to scale easily to similar analyses with much larger datasets. All steps here are able to run on an HPC cluster without modification, demonstrating reproducibility, automation, and capacity for high-throughput analyses.

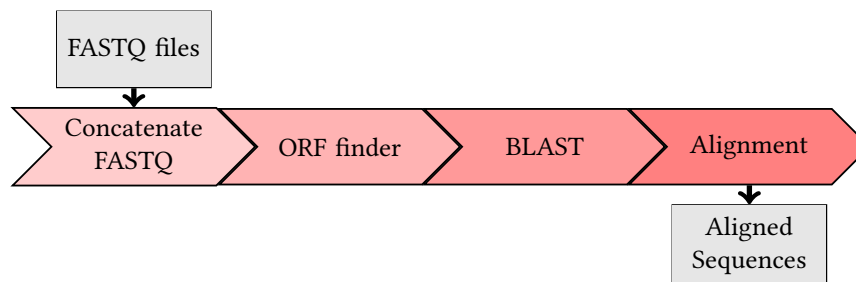


Figure 1: Workflow of Snakemake analysis pipeline.

Some python code:

```
from Bio import SeqIO
for record in SeqIO.parse("sequence.fasta", "fasta"):
    print(record.id, record.seq.translate())
```

Some Bash script:

```
#!/bin/bash
#SBATCH --job-name=blast_job
#SBATCH --partition=teach_cpu
#SBATCH --time=01:00:00
#SBATCH --mem=2000M

blastn -db nt -query sequence.fa -out results.out -outfmt 7 -remote
```

```
# BLASTP 2.17.0+
# Query: OMRGCv2f_11922010_1 [212 - 1072] [TARA OCEANS OM-RGCv2: [NPST] North Pacific
# Subtropical and Polar Front Provinces (MRGID:21484)]
# RID: G20BB3A3016
# Database: swissprot
# Fields: query acc.ver, subject acc.ver, % identity, alignment length, mismatches, gap opens,
# q. start, q. end, s. start, s. end, evalue, bit score
# 500 hits found
OMRGCv2f_11922010_1      A5UYW4.1      55.556      63      28      0      220      282
OMRGCv2f_11922010_1      A7NS65.1      55.556      63      28      0      220      282
OMRGCv2f_11922010_1      Q7NBW0.2      53.226      62      29      0      220      281
OMRGCv2f_11922010_1      B0BTI6.1      56.250      64      27      1      220      282
OMRGCv2f_11922010_1      A3N3J9.1      56.250      64      27      1      220      282
OMRGCv2f_11922010_1      Q9HJ83.1      55.385      65      27      1      220      282
OMRGCv2f_11922010_1      Q56237.2      50.000      62      31      0      220      281
OMRGCv2f_11922010_1      B9MJZ0.1      53.125      64      29      1      220      282
OMRGCv2f_11922010_1      O66921.1      51.613      62      30      0      220      281
OMRGCv2f_11922010_1      P50027.1      53.521      71      32      1      213      282
```

Query	Subject	% Identity	Alignment Length	E-value	Bitscore
OMRGCv2f_11922010_1	A5UYW4.1	55.56	63	1.3e-17	85.1
OMRGCv2f_11922010_1	A7NS65.1	55.56	63	1.8e-17	83.5

Table 1: Top 15 BLASTP hits for sample OMRGCv2f_11922010_1 against SwissProt.

Results

```
import pandas as pd
data = pd.read_csv("expression.csv")
normalized = data.apply(lambda x: np.log2(x+1))
```

Alignment results using Clustal Omega on some test data:

test1 . MGDVEKGKKIFIMKCSQCHTVEKGGKHKTGPNLHGLFGRKTGQAPGYSYTAANK 54
test2 . MGDVEKGKKIFVQKCAQCHTVEKGGKHKTGPNLHGLFGRKTGQAAGFSYTDANK 54
test3 KAGSAKKGATLFKTRCLQCHTVEKGGPHKVGPNLHGLFGRHSGQAEGYSYTDANI 59
consensus . MGDVEKGKKIF..KC.QCHTVEKGGKHKTGPNLHGLFGRKTGQA.GYSYTDANK

test1 NKGITWGEDTLMEYLENPKKYIPGTMIFVGIKKKEERADLIAYLKATNE 105
test2 NKGITWGEDTLMEYLENPKKYIPGTMIFAGIKKKGERADLIAYLKATNE 105
test3 KKNVLWDENNMSEYLTNPKKYIPGTMAFGLKKEKDRNDLITYLK..... 105
consensus NKGITWGEDTLMEYLENPKKYIPGTMIF.GIKKK.ERADLIAYLKATNE

☐ non-conserved
☐ ≥ 50% conserved
☐ ≥ 80% conserved

Figure 2: Alignment using Clustal Omega and Identity mode in texshade

Discussion

Yadadadada [1] and then also [2] and maybe even from this guy [3, 2]

References

- [1] WJ Kent. Blat—the blast-like alignment tool. *Genome research*, 12(4):656–664, 2002.
- [2] MJ Wolin, TL Miller, and CS Stewart. Microbe-microbe interactions. In *The rumen microbial ecosystem*, pages 467–491. Springer, 1997.
- [3] B Baker, P Zambryski, B Staskawicz, and SP Dinesh-Kumar. Signaling in plant-microbe interactions. *Science*, 276(5313):726–733, 1997.

Acknowledgments

This work was carried out using the computational facilities of the Advanced Computing Research Centre, University of Bristol - <http://www.bristol.ac.uk/acrc/>

Supplementary Data

Supplementary data can be found within the GitHub repository for this project: https://github.com/archiebenn/BIOLM0051_analysis.git